ICT2122 · Object Oriented Programming

Practical Tutorial: Design and Test a Small Model

Original synthetic reading for demonstration and independent study. It is not an official syllabus, lecture note, or assessment specification.

Learning objectives

After this reading, a learner should be able to explain the central idea, apply it to a small example, and identify at least one assumption or limitation.

1. Main idea

Start from a domain requirement, identify nouns and verbs, and assign each rule to the object that owns the relevant state. Keep constructors responsible for establishing invariants. Test both ordinary operations and rejected invalid values.

2. Example and reasoning

Model a course registration with Course, Student, and Enrollment objects. Check that an enrollment references one course and student, and that a duplicate registration is rejected. Replace storage with a fake when testing the model alone.

3. A method for practice

1. State the problem and define the terms that affect the answer.
2. Write down the assumptions and information available.
3. Work through a small example and make each step visible.
4. Check the result against a second case or independent explanation.
5. Summarize what the result supports and what it does not establish.

4. Questions for study

- Explain “practical tutorial: design and test a small model” in your own words and give a new example.
- Which assumption matters most in the example, and how would the answer change if it were false?
- Describe one common mistake and a practical way to detect it.
- Write a short checklist another learner could use to review this work.

Reflection

Separate evidence from assumptions. If a conclusion depends on information not provided here, identify what would need to be verified.